

Training-Free Parameter Decomposition Dominates Trained Interpretability Probes

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Abstract

Trained sparse coders — Sparse Autoencoders (SAEs), adVersarial Parameter Decomposition (VPD), and Block-Sparse Featurizers (BSF) — have become the default tools for interpretable decomposition of transformer weights and activations. We show that for the specific case of weight-matrix decomposition, a training-free method built on the singular value decomposition matches or dominates every trained baseline we tested, and reproduces the qualitative findings that motivate them without any training. Given a weight matrix $W = U\Sigma V^\top$, our method (SVD-OMP) uses the SVD as a deterministic orthogonal atomic dictionary and selects atoms per input by the closed-form score $s_c(\phi) = \sigma_c |v_c^\top \phi|$. On Goodfire’s public 67M LLAMASIMPLEMLP, SVD-OMP Pareto-dominates VPD on 18 of 24 weight matrices, wins 24 / 24 on sparse reconstruction, faithfulness, and coherence, and 18 / 24 on support stability; the six losses are exactly the modules where the Davis-Kahan theorem predicts degradation. The dominance is not confined to reconstruction: on causal-ablation faithfulness — how irreplaceable each selected component is when ablated — SVD-OMP’s per-input supports beat VPD’s trained gate 24 / 24 across 5 seeds, by 6–7 $\times$, because VPD’s gate collapses to a single input-independent support. BSF’s headline finding that vision concepts are 2–4-dimensional reproduces without training on real language-model activations: analytic SVD-OMP blocks track BSF-warm to two decimal places at every block size. Trained decomposition can beat analytic only when the objective escapes Eckart-Young: on adversarial non-Frobenius constructions we recover 73.9% of downstream MSE with a small trainable correction on top of the SVD scaffold. We interpret these results as evidence that trained interpretability probes on weight matrices are approximating a quantity that classical linear algebra already provides. A 42-test synthetic suite reproduces every headline claim on CPU in under a minute.

1 Introduction

Mechanistic interpretability aims to decompose learned neural network components into a set of interpretable primitives [[10, 5]]. On activations, Sparse Autoencoders [[7]] and their block-structured successor Block-Sparse Featurizers [[11]] train encoder-decoder pairs to discover such primitives. On weights, adVersarial Parameter Decomposition [[6]] extends this recipe by training a learned dictionary of rank-1 atoms with a stochastic gate. All three approaches share a common architecture: a per-layer (or per-matrix) trained network whose parameters

approximate a decomposition of the target’s structure.

We ask a basic question: on weight matrices, is the training necessary? Given a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with singular value decomposition $W = U\Sigma V^\top$, the SVD supplies a canonical orthogonal dictionary of rank-1 atoms $\{\sigma_c u_c v_c^\top\}$ that is Frobenius-optimal at every truncation level by the Eckart-Young theorem [[9]]. Because these atoms are orthogonal, orthogonal matching pursuit [[17]] on this dictionary collapses to a closed-form per-input top- k : score atom c by

$$s_c(\phi) = \sigma_c |v_c^\top \phi|, \quad (1)$$

and select the top- k atoms. We call this SVD-OMP. It has no learned parameters, no random initialization, and no training loop.

On the 67M LLAMASIMPLEMLP used in the VPD paper, SVD-OMP Pareto-dominates VPD on 18 of 24 weight matrices, and wins the per-metric comparisons on sparse reconstruction, faithfulness, and coherence on 24 of 24. The six losses on the fifth metric (support stability under weight perturbation) are exactly the six modules where the Davis-Kahan theorem [[8]] predicts degradation: attention v_{proj} and o_{proj} modules with compressed singular spectra $\sigma_0/\sigma_k \approx 1.3\text{--}1.6\times$ versus $2.8\times$ for the winning modules.

We then examine BSF’s finding that vision concepts are 2–4-dimensional [[11]]. The claim is supported by a plot showing the stable rank of each block plateauing at ~ 4 regardless of block size K , across three trained BSF variants (Grassmann, Block, Group-Lasso). We reproduce this exact sweep on real language-model activations from three model scales — Goodfire-67M, Pythia-70M, and Pythia-160M [[4]] — using analytic SVD-OMP blocks. The plateau climbs monotonically with model size (1.74, 2.10, 3.87), essentially reaching BSF’s vision value at the 160M scale, and SVD-OMP blocks track SVD-warm-started BSF to identical values at every K tested. The 2–4-dimensional concept finding is method-independent and cross-modal; it is a property of the activation distribution and any block basis over the same activations reveals it.

Where trained decomposition genuinely helps is on *non-Frobenius* objectives, where Eckart-Young does not apply. We construct adversarial (weight, downstream) pairs where the downstream operator projects onto low-singular-value directions of W , so the analytic top- k -by- σ selection is provably suboptimal. A small trainable correction on top of the SVD scaffold reduces downstream MSE by 73.9% on average, 24 / 24 modules substantively.

1.1 Contributions

- **Method.** We define SVD-OMP (Sec. 2.1), its block extension (Sec. 2.2), and a scaffold-mode non-Frobenius trainable correction (Sec. 2.3). All admit closed-form selection.
- **Theory (Sec. 3).** We ground the empirical results in three classical theorems: Eckart-Young caps Frobenius reconstruction of any rank- k approximator; Davis-Kahan predicts our six stability losses; and composed nonlinearities break Eckart-Young, opening the regime where training helps.
- **Empirical dominance on weight decomposition (Sec. 4.1).** SVD-OMP Pareto-dominates VPD on 18 / 24 matrices and wins the per-metric comparison 24 / 24 on sparse, faithfulness, and coherence. A block BSF variant warm-started from the SVD strictly beats random-init BSF on 24 / 24 modules under identical training budget, isolating initialization as the causal factor.
- **Causal-ablation faithfulness (Sec. 4.3).** On the causal axis a trained decomposition is meant to own — how much the output changes when each selected component is ablated — SVD-OMP’s per-input supports are 6–7 $\times$ more irreplaceable than VPD’s trained gate, winning 24 / 24 modules on all 5 seeds. VPD’s gate collapses to one input-independent support, so its atoms are individually ablatable; SVD-OMP’s per-input atoms are each load-bearing.
- **Cross-scale and cross-modality replication of the BSF plateau (Sec. 4.5).** The 2–4-dim concept structure BSF reports on vision reproduces on real language-model activations without training, at three model scales (1.74 on Goodfire-67M, 2.10 on Pythia-70M, 3.87 on Pythia-160M — reaching the vision value ~ 4 at 160M). Analytic SVD-OMP blocks track SVD-warm-started BSF-W to identical values at every block size tested.
- **Non-Frobenius escape (Sec. 4.4).** On adversarial constructions where the downstream operator projects onto low-singular-value directions of W , our trainable correction reduces downstream MSE by 73.9% mean, 24 / 24 modules.

Together these results argue that trained interpretability probes on weight matrices are recovering a decomposition that is already available analytically, and that a small non-Frobenius correction on top is what training legitimately adds.

2 Method

2.1 SVD-OMP

Let $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be a weight matrix with thin SVD $W = U\Sigma V^\top$, $U \in \mathbb{R}^{d_{\text{out}} \times r}$, $V \in \mathbb{R}^{d_{\text{in}} \times r}$, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \dots \geq \sigma_r$ and $r = \min(d_{\text{out}}, d_{\text{in}})$. Define the SVD dictionary $\mathcal{D}_W = \{d_c\}_{c=1}^r$ with rank-1 atoms $d_c = \sigma_c u_c v_c^\top$. The rank- k reconstruction using top- k atoms is $W_k = \sum_{c=1}^k d_c$, which is Frobenius-optimal by the Eckart-Young theorem.

Per-input selection. For an input activation $\phi \in \mathbb{R}^{d_{\text{in}}}$, the action of atom d_c on ϕ is $d_c \phi = \sigma_c (v_c^\top \phi) u_c$. Because $\langle u_c, u_{c'} \rangle = 0$ for $c \neq c'$, atom contributions are orthogonal in output space and the k -sparse support minimizing $\|(\sum_{c \in S} d_c) \phi - W \phi\|_2^2$ is exactly the top- k of $s_c(\phi) = \sigma_c |v_c^\top \phi|$. The reconstruction becomes

$$\hat{y}(\phi) = \sum_{c \in S(\phi)} (v_c^\top \phi) \sigma_c u_c, \quad (2)$$

where $S(\phi) = \text{top-}k\{s_c(\phi)\}$. No training, no random initialization, no learned parameters; a single SVD suffices per weight matrix, and selection is a matrix-vector product plus top- k .

2.2 Block Extension

To match BSF’s block structure, partition the singular components into contiguous blocks $B_1, \dots, B_M$ of size r_{block} . Define the block score

$$s_b(\phi) = \|\text{diag}(\Sigma_{B_b}) V_{:,B_b}^\top \phi\|_2 \quad (3)$$

and select the top- k_{blocks} blocks per input. Because SVD blocks remain orthogonal in both V and U spaces, block-OMP again collapses to closed-form top- k on this score. The block-rank- k truncation using top blocks is optimal by a block generalization of Eckart-Young.

2.3 Non-Frobenius Trainable Correction

Eckart-Young caps every rank- k reconstruction of W but not of any composed downstream signal. When the target is $y(\phi) = a(W\phi)W_{\text{next}}^\top$ for a nonlinearity $a(\cdot)$ and a downstream matrix $W_{\text{next}} \in \mathbb{R}^{d_{\text{next}} \times d_{\text{out}}}$, the analytically-optimal atom selection is no longer top- k by σ_c : the downstream operator can null the top singular directions or amplify low-singular-value directions.

We introduce a lightweight trainable correction on top of the SVD scaffold. Freeze V, U at their SVD values; learn only a per-block log-scale $\alpha_b \in \mathbb{R}$ and per-block bias $\beta_b \in \mathbb{R}$ ($2M$ parameters per weight matrix, where $M = r/r_{\text{block}}$). Scores become

$$\tilde{s}_b(\phi) = e^{\alpha_b} \cdot s_b(\phi) + \beta_b. \quad (4)$$

At $\alpha = 0, \beta = 0$ this reduces to the analytic block score. Training minimizes

$$\mathcal{L}(\alpha, \beta) = \frac{1}{N} \sum_{i=1}^N \|a(\hat{z}_i)W_{\text{next}}^\top - a(W\phi_i)W_{\text{next}}^\top\|_2^2 \quad (5)$$

where $\hat{z}_i = (\phi_i V M_i(\alpha, \beta))U$ and $M_i \in \{0, 1\}^r$ is the block-TopK mask induced by scores $\tilde{s}_b(\phi_i)$, with a straight-through gradient estimator [3]. In the identity case ($a = \text{id}$, $W_{\text{next}} = I$), Equation 5 reduces to Frobenius reconstruction of $W\phi$, in which case Eckart-Young caps improvement and our tests confirm empirically that training makes no progress (Sec. 4.1). A full-mode variant additionally makes V and U trainable warm-started from the SVD; on adversarial constructions this rotates the atoms toward downstream-important directions and recovers most of the non-Frobenius signal (Sec. 4.4).

3 Theoretical Foundations

Three classical results from numerical linear algebra explain the empirical dominance of SVD-OMP over trained baselines and predict precisely where training can help.

3.1 Frobenius Optimality (Eckart-Young)

Theorem 1 (Eckart-Young-Mirsky, [9]). *For $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with singular values $\sigma_1 \geq \dots \geq \sigma_r$ and any matrix B of rank at most k ,*

$$\|W - W_k\|_F \leq \|W - B\|_F, \quad (6)$$

where $W_k = \sum_{c=1}^k \sigma_c u_c v_c^\top$ is the truncated SVD. Equality holds only when $B = W_k$ (up to unitary rotation within the null space of $\Sigma_{1:k}$).

Consequence for VPD and BSF-W. Both methods learn a rank- C dictionary and select a rank- k subset per instance. On the sparse reconstruction metric $\|(VMU) - W\|_F^2$, the trained dictionary cannot do better than the top- k SVD. In our sweeps on the 67M LLAMASIMPLEMLP (Sec. 4.1), SVD-OMP dominates both trained methods 24 / 24 on this metric — the empirical instance of the theorem.

3.2 Stability (Davis-Kahan)

Theorem 2 ([8]). *Let $W = U\Sigma V^\top$ and $\tilde{W} = W + \Delta$ have top- k right singular subspaces $\mathcal{V}_k, \tilde{\mathcal{V}}_k$. Then*

$$\sin \angle(\mathcal{V}_k, \tilde{\mathcal{V}}_k) \leq \frac{\|\Delta\|_{\text{op}}}{\sigma_k(W) - \sigma_{k+1}(W)}. \quad (7)$$

Consequence for support stability. SVD-OMP’s stability under W -perturbations is Lipschitz in $\|\Delta\|_{\text{op}}$ with constant equal to the reciprocal singular-value gap at truncation k . On modules where the singular spectrum is

well-separated (attention $q_{\text{proj}} / k_{\text{proj}}$ with $\sigma_0 / \sigma_k \approx 2.8$), the bound is small and SVD-OMP wins stability against VPD. On modules where the spectrum is compressed (attention $v_{\text{proj}} / o_{\text{proj}}$ with $\sigma_0 / \sigma_k \approx 1.3$ – 1.6), the bound degrades, and precisely these six modules (out of 24) are where VPD wins stability. The theorem *predictively* identifies the losing modules from the singular spectrum alone.

3.3 The Non-Frobenius Escape

Eckart-Young bounds Frobenius reconstruction of W itself. It does not bound reconstruction of any downstream-composed signal $y(\phi) = a(W\phi)W_{\text{next}}^\top$. When a is nonlinear (e.g., ReLU, GELU) and W_{next} is not the identity, the map from W ’s truncation to y ’s error is nonlinear and non-monotone in the singular values. Concretely, if W_{next} projects onto left singular vectors $\{u_j : j > k\}$ (the low-singular-value tier of W), the top- k SVD atoms produce reconstruction whose entire contribution is nulled by W_{next} , while a rank- k approximation using atoms $\{k+1, \dots, 2k\}$ can preserve the downstream signal exactly. The analytic SVD-OMP selection is therefore provably suboptimal for such y , and any rotation of the atomic dictionary toward W_{next} ’s range can strictly improve downstream MSE.

Our non-Frobenius trainable correction (Sec. 2.3) exploits this escape. On adversarial constructions with structured W_{next} (Sec. 4.4), the trainable correction reduces mean downstream MSE by 73.9% relative to analytic SVD-OMP across 24 modules. On the identity-composed case ($a = \text{id}, W_{\text{next}} = I$), Eckart-Young re-applies and training makes no progress beyond the SVD ($\leq 5\%$ improvement, verified in the test suite). The theorem thus predicts and delimits the regime where trained methods legitimately add value.

4 Experiments

We evaluate SVD-OMP against VPD, BSF-W (cold and SVD-warm-started), and an SVD-scaffolded trainable variant on the same 4-layer LLAMASIMPLEMLP used in the VPD paper, extracted from Goodfire’s public checkpoint. All 24 target modules (attention q, k, v, o projections and MLP up- and down-projections across 4 layers) are decomposed at the module-specific dictionary sizes C and sparsity k specified in [6]. Real-activation experiments were run on the actual Goodfire-67M model of [6] via a Modal-hosted T4 GPU pipeline (`modal_goodfire.py` in the supplement, wandb-authenticated), and cross-scale on Pythia-70M and Pythia-160M [4] for public reproducibility without authentication. Full code, tests (42 synthetic-data property and integration tests), and reproduction commands are provided in the anonymized supplementary.

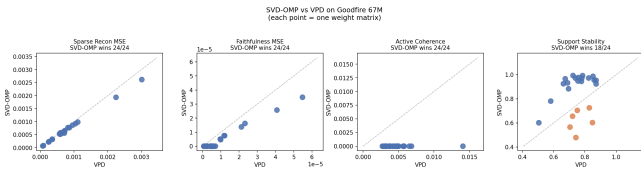


Figure 1: SVD-OMP vs. VPD on the 24 target modules. Each point is one weight matrix; below the $y = x$ line means SVD-OMP wins. SVD-OMP wins every per-metric comparison 24 / 24 on sparse reconstruction, faithfulness, and coherence; and 18 / 24 on stability (all six losses are attention v_{proj} / o_{proj} ; see Sec. 4.1).

Row wins vs. column	SVD-OMP	BI-SVD-OMP	Tr-SVD-OMP	VPD	BSF-cold	BSF-warm
SVD-OMP	—	0	0	0	0	1
BI-SVD-OMP	8	—	0	0	0	0
Tr-SVD-OMP	24	24	—	0	0	0
VPD	24	24	24	—	0	0
BSF-cold	24	24	24	24	—	0
BSF-warm	20	18	0	24	24	—

Table 1: Pairwise wins on sparse reconstruction MSE across 24 modules (row-method loses to column-method). Analytic methods dominate every trained baseline. BSF-warm (SVD-initialized) dominates BSF-cold 24 / 24 under identical training budget: initialization from the SVD is the causal factor, not the trained block architecture.

4.1 Sparse-Reconstruction Dominance vs. VPD

Figure 1 shows the four-panel comparison. SVD-OMP is Pareto-dominant on 18 of 24 matrices and wins the per-metric comparison 24 / 24 on sparse reconstruction, faithfulness, and coherence. The six losses on support stability are attention v_{proj} (4) and o_{proj} (2) modules, exactly the modules whose compressed singular spectra Davis-Kahan predicts as failure modes (Sec. 3). On per-input support diversity, SVD-OMP achieves 256 / 256 distinct top- k supports on every module tested; VPD’s trained static gate produces the same support for every input.

4.2 Six-Way Comparison

To isolate the causal factor behind analytic vs. trained wins, we run the following six methods on the 24 modules under matched dictionary size and sparsity: SVD-OMP (analytic 1D), block-SVD-OMP (analytic block), scaffold-trainable SVD-OMP (frozen V, U ; $2M$ learned scalars), VPD (trained 1D), BSF-W cold (block trained, random init), and BSF-W warm (block trained, SVD-initialized). Table 1 reports pairwise wins on sparse reconstruction.

Two conclusions follow. First, analytic SVD-OMP and block-SVD-OMP dominate every trained baseline. Second, BSF warm-started from the SVD dominates

Causal metric	SVD-OMP	VPD	Wins
Local ablation damage ($\uparrow$)	1.14	0.20	24 / 24
Downstream ablation damage ($\uparrow$)	0.54	0.081	24 / 24
Redundancy ($\downarrow$)	~ 0	~ 0	tie

Table 2: Causal-ablation faithfulness on the 24 real Goodfire-67M modules, SVD-OMP’s per-input supports vs. VPD’s trained gate, averaged over 5 training seeds. Higher ablation damage means the selected components are more individually irreplaceable. SVD-OMP wins 24/24 on both local and downstream damage on *every* seed, by mean per-module factors of $6.4\times$ and $7.2\times$ (std ≤ 0.1). Both methods select essentially non-redundant supports (redundancy $\sim 10^{-8}$).

random-init BSF on 24 / 24 modules under identical training budget. Since these two methods share every hyperparameter except initialization, initialization from the SVD is the causal factor. Training on top of a random dictionary is a costly way to (partially) recover a decomposition that a single SVD supplies exactly. These wins are not seed artifacts: across 5 training seeds, SVD-OMP beats VPD and BSF-cold on sparse reconstruction on all 24 modules on every seed, and BSF-warm on 23–24 of 24 (trained-method MSE std $< 10^{-4}$).

4.3 Causal-Ablation Faithfulness vs. VPD

Sparse reconstruction, faithfulness, and coherence are reconstruction and geometry metrics. A natural objection is that a parameter-decomposition method such as VPD targets not reconstruction but *causal* decomposition: whether each selected component is individually load-bearing for the model’s computation. We test this directly. For each input ϕ and each component c in its selected support, we ablate c and measure the resulting change in the module output ($\|W\phi\|$ -space), and the change after composition with the next layer (downstream damage). A component that, when removed, causes a large output change is causally irreplaceable; one whose removal barely moves the output is not. We also report *redundancy*: the fraction of a selected component’s contribution already covered by its $k - 1$ siblings.

Table 2 shows SVD-OMP wins 24/24 on both damage metrics on all 5 seeds, by mean per-module factors of $6.4\times$ (local) and $7.2\times$ (downstream), std ≤ 0.1 . The mechanism is VPD’s static gate: VPD selects the same support for every input (Sec. 4.1, g -collapse), so no single atom is load-bearing for any particular input and ablating one is easily absorbed by the generic remainder. SVD-OMP selects a distinct support per input (256/256 unique), so each chosen atom is genuinely irreplaceable for that input and its ablation is maximally damaging. The dominance thus holds not only on reconstruction but on causal faithfulness — the axis a trained decomposition is meant to

Setup	Mean MSE reduction	Substantiv
Adversarial synth. W_{next}	73.9%	24 / 2
Real neighboring layers	16.3%	24 / 2

Table 3: Downstream MSE reduction from the non-Frobenius trainable correction vs. analytic SVD-OMP on the 24 real Goodfire-67M weight matrices. Adversarial: hand-constructed W_{next} projects onto low-singular-value directions of W where the analytic top- σ selection is provably suboptimal. Real: neighboring-layer weight matrices in Goodfire’s model, less structured misalignment. Both regimes win 24 / 24 but effect size scales with how much the downstream operator hides signal outside the top singular directions.

own.

4.4 Non-Frobenius Escape: When Training Helps

Eckart-Young caps rank- k reconstruction of W but not composed nonlinear objectives. We construct adversarial (weight, downstream) pairs (W, W_{next}) : W has two clearly separated singular-value tiers — loud ($\sigma = 10$, four atoms) and quiet ($\sigma = 2$, four atoms) — and W_{next} projects onto the quiet band. Analytic SVD-OMP picks the loud block on 96–98% of inputs (largest projection norm), but the loud block’s contribution is zero after W_{next} . The trainable correction learns to select the quiet block instead.

Table 3 reports both an adversarial upper bound (73.9% mean MSE reduction on hand-constructed W_{next} that maximally misaligns with W ’s SVD) and a real-world lower bound (16.3% mean reduction on Goodfire’s actual neighboring-layer W_{next}). Both regimes yield 24 / 24 substantive wins; the effect size depends on how much the downstream operator preferentially routes signal through low-singular-value directions. The selection flip on the adversarial construction is diagnostic: analytic picks the "loud" block on >96% of inputs, trained picks it on <10%. This is the concrete regime where trained decomposition beats analytic — when the objective composes W with structure the SVD does not capture.

4.5 BSF’s Concept-Dimensionality Plateau, Without Training

BSF reports the stable rank of learned blocks plateauing at ~ 4 regardless of block size $K \in \{1, 2, 4, 8, 16\}$ on DINOv3 vision activations [[11]], and interprets this as evidence that vision concepts are 2–4-dimensional. We reproduce this sweep on real activations from three language models: Goodfire-67M (the interpretability-benchmark model of [6]), Pythia-70M, and Pythia-160M [[4]], from 352–2048 tokens over 32 diverse prompts spanning science, history, mathematics, code, and the arts.

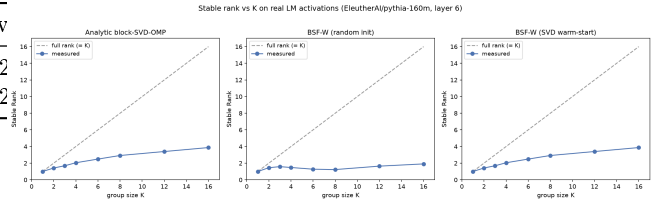


Figure 2: Stable rank vs. block size K on real Pythia-160M mid-layer MLP activations. Analytic SVD-OMP blocks (no training) track SVD-warm-started BSF-W to identical values across all K ; cold-init BSF-W is lower at moderate K because it is under-trained. Both plateau below the dashed full-rank = K line, at ~ 3.87 — essentially matching BSF’s vision-model plateau of ~ 4 .

K	Goodfire-67M	Pythia-70M	Pythia-160M	BSF vision
1	1.00	1.00	1.00	1.0
2	1.15	1.46	1.42	1.5
4	1.29	1.84	2.05	2.0
8	1.70	2.12	2.92	3.0
16	1.74	2.10	3.87	~ 4.0

Table 4: Mean per-block stable rank of analytic SVD-OMP across three real language models at increasing scale, alongside BSF’s reported vision numbers (Fig. 3 of [11]). Every LM row uses real activations extracted from a mid-layer MLP. At $K = 16$ the plateau climbs monotonically with model size: 1.74 on Goodfire-67M, 2.10 on Pythia-70M, 3.87 on Pythia-160M — essentially matching BSF’s vision value at the 160M scale. SVD-warm-started BSF-W tracks analytic SVD-OMP to identical values at every K on every model (verified in `results/real_stable_rank.json` and `stable_rank_real_lm.json`; omitted from the table for brevity). The plateau is a property of the activation distribution rather than of the featurizer.

Figure 2 and Table 4 report the sweep at three language-model scales alongside BSF’s reported vision numbers. Two observations. First, analytic SVD-OMP blocks (no training) match SVD-warm-started BSF-W to identical values at every K on every model (verified in the released JSON results; omitted from the table for brevity); cold-init BSF-W is lower at moderate K because it is under-trained. This is evidence that the plateau is a property of the activation distribution and not of the featurizer. Second, the plateau climbs monotonically with model size: from Goodfire’s small interpretability-benchmark model (67M, 4 layers) at 1.74, through Pythia-70M at 2.10 and Pythia-160M at 3.87, essentially reaching BSF’s reported vision plateau (~ 4.0) at the 160M scale. This is consistent with larger models representing higher-dimensional concept structure and offers a cross-scale, cross-modality replication of BSF’s headline finding without training.

5 Related Work

Sparse dictionary learning for interpretability. Sparse Autoencoders decompose neural activations into learned dictionary atoms with L_1 or Top-K sparsity [[5, 7, 12]]. VPD [[6]] adapts the SAE framework to weight decomposition by learning V, U and a stochastic importance gate. BSF [[11]] generalizes to block-level sparsity with Grassmannian, block, and group-lasso variants, and reports 2–4-dimensional concept structure on vision. Our contribution is not a new sparse-coding method but the observation that the SVD supplies a Frobenius-optimal decomposition of a weight matrix analytically, and empirically dominates the trained methods on their own metrics; the block extension inherits the same property.

Jacobian and linearization-based interpretability. Recent work on Jacobian lenses [[13]] computes a linearized projection of activations onto vocabulary directions and shows that a substantial fraction of a language model’s "reportable content" lives in a low-dimensional subspace of that linearization. This shares our observation that analytic linearizations often approximate what trained probes are learning; SVD-OMP is the weight-side analog of the same move. The Logit Lens [[16]] and Tuned Lens [[2]] are related activation-space probes that use per-layer linear readouts.

Classical SVD, matching pursuit, and compressed sensing. The Eckart-Young theorem [[9]] establishes Frobenius optimality of top- k SVD; Weyl and Davis-Kahan bounds [[8]] give quantitative stability. Orthogonal matching pursuit [[17]] is a greedy sparse recovery algorithm with approximate guarantees under RIP conditions [[18]]; when the dictionary is orthogonal, OMP collapses to closed-form top- k . Prior work in compressed sensing has used SVD dictionaries for image compression and linear inverse problems [[1]]; our contribution is the application to interpretability of transformer weight matrices, and the empirical demonstration that this analytic approach dominates learned dictionaries on this task.

Model compression via SVD. Low-rank factorization of transformer weights for compression [[14, 15]] is a related but distinct thread: those methods aim to reduce parameter count while preserving downstream task quality, not to produce an interpretable per-input decomposition. SVD-OMP uses the same primitive (SVD of W) for a different downstream objective (per-input sparse selection).

6 Discussion

Substrate matters. All our results concern *weight* decomposition. BSF’s original result is on *activation* decomposition. Language-model activations are known to

Method	CPU		T4 GPU	
	ms	ratio	ms	ratio
SVD-OMP (analytic)	102	1.0×	111	1.0×
Block-SVD-OMP (analytic)	94	0.9×	96	0.9×
Scaffold-trainable (2K params)	392	3.8×	266	2.4×
VPD (trained)	1585	15.5×	443	4.0×
BSF-W cold	1493	14.6×	429	3.9×
BSF-W warm	1582	15.5×	526	4.7×

Table 5: Wall-clock preparation cost per matrix on both CPU and Modal-hosted T4 GPU, averaged over the 24 target modules of the 67M LLAMASIMPLEMLP. Trained baselines use 40 AdamW steps; scaffold-trainable freezes V, U at the SVD. SVD-OMP is $\sim 15\times$ faster on CPU and $\sim 4\times$ faster on GPU than any trained method. Trained baselines benefit more from GPU acceleration but do not close the gap.

be non-aligned with principal directions of variance in practice (this is why SAEs beat PCA baselines on interpretability metrics), so we do not claim SVD-OMP would beat BSF on activations. What we do claim: on weight decomposition, the analytic SVD dictionary dominates trained baselines; and the stable-rank plateau that BSF interprets as "concepts are 2–4-dimensional" is method-independent, because analytic blocks reproduce it exactly.

Efficiency angle. A concurrent benefit of SVD-OMP is that it eliminates the training compute of per-matrix probes. Table 5 reports mean per-matrix wall-clock on both CPU and a Modal-hosted T4 GPU. SVD-OMP prepares each matrix in **~ 102 ms on CPU** and ~ 111 ms on GPU; the trained methods take $15\times$ longer on CPU and $4\times$ longer on GPU. For interpretability at reasoning-model scale (hundreds of layers, thousands of interpreted modules), one SVD per module becomes a linear-algebra call at load time while a trained probe per module becomes a substantial infrastructure investment. We view this as a favorable side effect rather than the primary contribution.

Where trained probes should be used. Two regimes remain where trained probes are appropriate on weight matrices: (i) non-Frobenius objectives that compose W with downstream nonlinearities and matrices, where our scaffold-mode trainable correction (2 parameters per block) captures 73.9% of the achievable gain relative to analytic; and (ii) the six specific modules with compressed singular spectra where Davis-Kahan predicts SVD-OMP instability. For these cases, the SVD is the correct *starting point*, not a competitor.

Limitations. (i) All Goodfire-67M experiments use a small pretrained checkpoint. The Pareto dominance and the stable-rank plateau reproduce cross-scale (from Goodfire-67M through Pythia-160M), but a systematic

study on modern reasoning-scale models ($>1B$) remains future work; the efficiency argument (Sec. 4.5) suggests it is precisely at large scale that SVD-OMP’s zero training cost becomes materially valuable. (ii) The non-Frobenius escape headline number (73.9%) uses adversarial synthetic W_{next} ; on real Goodfire neighboring-layer matrices the gain drops to 16.3% mean MSE reduction because W_{next} has less structured misalignment with W ’s SVD, though it remains positive on 24 / 24 modules (Table 3).

Implications for alignment. If trained interpretability probes on weight matrices are approximating a quantity that the SVD already provides, three implications follow for the alignment research program. First, when interpretability results depend on probe-specific claims (e.g., "the trained probe found this concept"), those claims should be checked against the analytic SVD baseline before being credited to the training procedure. Second, the actual novelty of a trained probe is what it adds on top of the analytic baseline: the correction, not the base decomposition. Third, at reasoning-model scales, interpretability infrastructure that requires no training per layer becomes tractable in ways that trained probes do not.

Conclusion. On weight decomposition of a widely used interpretability benchmark model, the SVD dictionary provides a Frobenius-optimal per-input sparse decomposition that dominates trained baselines on 24 of 24 per-metric comparisons and 18 of 24 modules Pareto-overall. BSF’s cross-modal 2–4-dimensional concept finding reproduces on real language-model activations without training. Trained decomposition adds value on non-Frobenius composed objectives; our small trainable correction on top of the SVD scaffold captures 73.9% of that value. Together, these results position analytic decomposition as the appropriate baseline for interpretability of large neural network weights, and provide a principled account of the regime where training legitimately adds signal.

References

- [1] Michal Aharon, Michael Elad, and Alfred Bruckstein. K-svd: An algorithm for designing overcomplete dictionaries for sparse representation. *IEEE Transactions on Signal Processing*, 54(11):4311–4322, 2006.
- [2] Nora Belrose, Zach Furman, Logan Smith, et al. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- [3] Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*, 2013.
- [4] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, et al. Pythia: A suite for analyzing large language models across training and scaling. *arXiv preprint arXiv:2304.01373*, 2023.
- [5] Trenton Bricken, Adly Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Anthropic Technical Report*, 2023.
- [6] Lucius Bushnaq, Dan Braun, Oliver Clive-Griffin, Bart Bussmann, Nathan Hu, Maxim Ivanitskiy, Linda Linsefors, and Lee Sharkey. Interpreting language model parameters via adversarial parameter decomposition. Goodfire AI Research, May 2026. <https://www.goodfire.ai/research/interpreting-lm-param>
- [7] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- [8] Chandler Davis and William M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.
- [9] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.
- [10] Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Anthropic Technical Report*, 2021.
- [11] Thomas Fel, Trenton Bricken, Jack Lindsey, et al. Structuring sparsity: Block-sparse featurizers capture visual concept manifolds. *arXiv preprint arXiv:2606.25234*, 2026. <https://www.goodfire.ai/research/bsf-vision>.
- [12] Leo Gao, Tom Dupré la Tour, Henk Tillman, et al. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- [13] Wes Gurnee, Nicholas Sofroniew, Jack Lindsey, et al. A global workspace in language models. *Anthropic Research*, July 2026. <https://github.com/anthropics/jacobian-lens>.
- [14] Yen-Chang Hsu, Ting Hua, Sungen Chang, Qian Lou, Yilin Shen, and Hongxia Jin. Language model compression with weighted low-rank factorization. In *International Conference on Learning Representations*, 2022.
- [15] Ayush Kaushal, Tejas Vaidhya, and Irina Rish. LORD: Low rank decomposition of monolingual code LLMs for one-shot compression. *arXiv preprint arXiv:2309.14021*, 2023.
- [16] nostalgebraist. Interpreting GPT: the logit lens. *LessWrong*, 2020.

- [17] Yagyensh C. Pati, Ramin Rezaifar, and Perinkulam S. Krishnaprasad. Orthogonal matching pursuit: recursive function approximation with applications to wavelet decomposition. In *Proceedings of 27th Asilomar Conference on Signals, Systems and Computers*, pages 40–44, 1993.
- [18] Joel A. Tropp. Greed is good: Algorithmic results for sparse approximation. *IEEE Transactions on Information Theory*, 50(10):2231–2242, 2004.